

EE446-TinyML Lab 9

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Github: <https://github.com/sdadhich04/EE-446-TinyML-/tree/main/Lab-9>

```
hi: 0.932795  
sup: 0.067205
```

```
hi: 0.903402  
sup: 0.096598
```

```
hi: 0.910415  
sup: 0.089585
```

```
hi: 0.941554  
sup: 0.058446
```

```
hi: 0.966912
```

```
hi: 0.293987  
sup: 0.706013
```

```
hi: 0.330109  
sup: 0.669891
```

I found and fixed an issue in the notebook where the model was compiled using `loss='mse'` and `metrics=['mae']`, which are intended for regression tasks rather than multi-class classification. Since this model uses a softmax output layer to classify ASL gestures, I changed the loss function to `categorical_crossentropy` and the evaluation metric to `accuracy`, which are more appropriate for this task. To keep the notebook consistent with this change, I also updated the metric plotting variable from "mae" to "accuracy" so that the plots, labels, titles, legends, and training history references correctly displayed classification performance. After making these changes, I retrained the model, generated the updated evaluation plots, quantized the model, and exported it as `model.h` for deployment on the Arduino board.